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Lightweight Cryptography: A Solution to Secure IoT

Sumit Singh Dhanda¹ · Brahmjit Singh¹ · Poonam Jindal¹

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Abstract

In Internet of Things (IoT), the massive connectivity of devices and enormous data on the air have made information susceptible to different type of attacks. Cryptographic algorithms are used to provide confidentiality and maintain the integrity of the information. But small size, limited computational capability, limited memory, and power resources of the devices make it difficult to use the resource intensive traditional cryptographic algorithms for information security. In this scenario it becomes impertinent to develop lightweight security schemes for IoT. A thorough study on the lightweight cryptography as a solution to the security problem of resource-constrained devices in IoT has been presented in this work. This paper is a comprehensive attempt to provide an in-depth and state of the art survey of available lightweight cryptographic primitives till 2019. In this paper 21 lightweight block ciphers, 19 lightweight stream ciphers, 9 lightweight hash functions and 5 variants of elliptic curve cryptography (ECC) has been discussed i.e. in total 54 LWC primitives are compared in their respective classes. The comparison of the ciphers has been carried out in terms of chip area, energy and power, hardware and software efficiency, throughput, latency and figure of merit (FoM). Based on the findings it can be observed that AES and ECC are the most suitable for used lightweight cryptographic primitives. Several open research problems in the field of lightweight cryptography have also been identified.

Keywords Elliptic curve cryptography (ECC) · Internet of Things (IoT) · Lightweight cryptography · Lightweight block ciphers · Lightweight stream ciphers · Lightweight hash functions

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1 Introduction

IoT can be considered as the first evolution of the Internet [1]. Internet of Things (IoT) is a ubiquitous network [2] of uniquely identifiable things, real or virtual, that communicates massive amount of data to be used for intelligent decision making. With IoT, the number of things connected to internet is expected to grow up to 50 billion by 2020. IoT systems provide a large spectrum of services such as Intelligent Transportation Systems (ITS), smart grids, smart buildings, smart cities, e-Health, intelligent drug delivery system etc. Even the Cyber-Physical Systems (CPS) such as Nuclear Power Plant (NPP) comes under IoT umbrella. Most of these services are critical in nature.

Every IoT system is designed to provide a specific service. Delivery of the services depends upon the information collected at the perception layer. It is the lower-most layer in IoT. Resource-constrained devices or Wireless Sensor Networks (WSNs) constitute the perception layer. Most of these devices are openly deployed and use wireless media for transmission. Open deployment makes these devices susceptible to node tampering. It is also quite easy to intercept a wireless transmission and alter the content of information. Given the critical nature of services, information security is of paramount importance as the interception or alteration of the information can result in heavy loss of life and money.

Dave Evans [1] mentions that in an IoT application for cow monitoring, on average each cow generate 200 mega-bytes (MB) of information every year. A connected car generates tens of mega-bytes of data per second. An autonomous vehicle can generate up to 1 gigabyte of data per second. Considering 50 billion devices and the amount of data generated per device, a huge amount of data is placed in the air making it vulnerable to different kind of attacks. These attacks are increasing day by day. Denial of service, man-in-the-middle, zero day, identity theft, and malware are some crucial attacks. Denial of service attacks can prove fatal in e-Health and ITS. Similar attack can paralyze a whole city in case of smart cities and NPPs. Identity theft can pose threat to a nation's security or financial loss to an organization. According to Internet Security Threat Report (ISTR) 2019 by Symantec [3] crypto-jacking attacks were around 4 million in number in December 2018 and around 4800 websites were compromised with form-jacking attacks. Highest contributors of attacks on IoT devices were worms and bots. Proof of concept attacks on IoT products by Ronen and Shamir [4] demonstrates their vulnerability. A smart hacker can use these vulnerabilities [4–9] to attack various IoT devices such as light-bulbs, smart TVs, smart tablets etc. In [10] authors have shown the vulnerability of cross-protocol IoT products using hierarchical attack representation models. These attacks underline the need of developing new security mechanisms for IoT.

Many organizations and research agencies have stressed upon the need of security for IoT. Open Web Application Security Project (OWASP) [11] has identified privacy, insufficient authentication/authorization, lack of transport encryption, and poor physical layer security among the top ten vulnerabilities for IoT. As per IoT reference architecture, IoT security has five functional components [12], identity management, authentication, authorization, key exchange and management, trust, and reputation. Figure 1 shows the major thrust area in the field of IoT security. It includes authentication, access control non-repudiation apart from the confidentiality, integrity, and availability. With the help of cryptographic primitives all of these objectives can be fulfilled.

Confidentiality and integrity of the information can be achieved by cryptography. But traditional cryptographic methods require large allocation of resources. On the other hand, IoT devices are characterized [13] by limited computational power, limited memory,

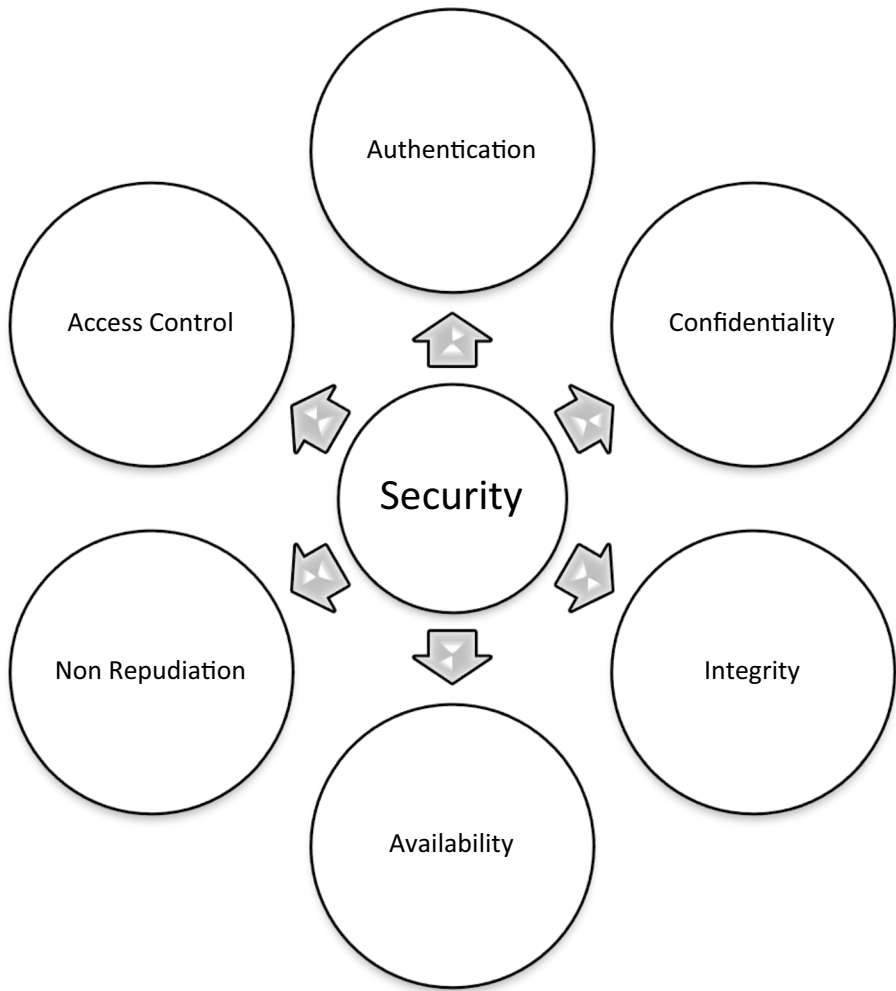


Fig. 1 Thrust area in security

limited power supply, and limited battery life. In [14] authors have compared different sensor motes used for the WSN and found that resource-constrained devices have as low as 2 kilo bytes (kB) and 1 kB of Random Access Memory (RAM) and Electrically Erasable Programmable Read Only Memory (EEPROM) respectively. Such sensors can't utilize the resource-consuming traditional security methods. Their qualities are in complete contrast to the security requirements in these networks. Hence, security is one of the main challenges in low power and lossy networks [15], [16]. This clearly outlines the need to develop Lightweight Cryptographic (LWC) algorithms for information security.

In this paper, current state of the art in the field of lightweight cryptography has been presented covering the time line till 2019. A comparative evaluation of existing and some of the newly proposed lightweight ciphers such as SFN, Espresso, Fruit-v2, and Lizard etc. has been presented. An extensive performance evaluation has been carried out in terms of chip area, energy, and power, hardware and software efficiency, throughput latency and

Figure of Merit (FoM). FoM given by Badel et al. [17] is used for the first time in a survey to compare different ciphers. It is an important parameter which removes the effect of CMOS technology on the ‘throughput’ of the cipher. After the detailed examination research gaps have been identified. This paper has been organised into nine parts. Section I introduces IoT and the need for the development of LWC. Section II presents current state of the art and other related works. In Section III classification and applications of cryptographic primitives have been presented. Section IV-VII explains and compares the crucial Lightweight Block Ciphers (LWBCs), Lightweight Stream Ciphers (LWSCs), Lightweight Hash Functions (LWHFs) and Elliptic Curve Cryptography (ECC) respectively. In Section VIII different parameters to evaluate a LWC primitive have been discussed at length and the research gaps are provided. Based on individual parameter a detailed comparison of the mentioned LWC primitives have been plotted. Finally, conclusion is provided in Section IX.

2 Related Work and Contribution

In [13], authors have elaborated various aspects of the lightweight cryptography. Authors have purposed a lightweight hybrid algorithm for the IoT devices. It tells which LWC algorithm should be used on a specific device. This decision is made on the basis of memory storage, and power of the device alongside the computational power required for the LWC algorithm. This article covered the timeline until 2016. In [14], a thorough evaluation of different block ciphers has been carried out and their performance has been compared. Authors have also provided the details of sensor motes used in WSNs. In [18], authors have studied eight different IoT frameworks and presented the requirements to develop third party applications for IoT systems. Authentication and access control are the primary security need for any IoT application and framework. This work helps the developers to improve the security and design of their systems and application.

In [19] a detailed assessment of lightweight block ciphers and their implementations have been submitted. It has been shown that the CMOS technology has an impact upon the gate equivalence required for the hardware implementation of a cipher. In [20] a tutorial on the cryptographic primitives has been presented. It also emphasises the need of the lightweight algorithms. In [21], symmetric ciphers for the resource-constrained environment have been studied and evaluated. A detailed survey on the lightweight block ciphers for low resource devices, that covers the scenario till 2013 has been provided in [22].

Aziz and Singh [23] have presented compressive sensing as a means to provide lightweight security for the IoT. In this work compressive sensing has been used for encryption of the data. It helps in conserving the energy. In [24], the need for the security for the low power and lossy networks has been identified. It has outlined the deficiency of Datagram Transport Layer Security (DTLS) and Internet Protocol version 6 (IPv6) over wireless personal area network (6LoWPAN). A new protocol for the security of IoT has been proposed which ensures end-to-end security of 6LoWPAN. A new protocol for the security of IoT has been proposed in [25] which utilizes random and separate key for encryption of every file. It also uses probabilistic encryption to avoid chosen plaintext attacks (CPA). Authors have compared it with Internet Protocol Security (IPSec) and found it better in performance. A detailed tutorial has been presented in [26] on Physically Unclonable Functions (PUF). These functions are based on special characteristics of the Integrated Chips (ICs)

to exhibit the unique pattern and response to a given stimulus. In this manner ICs can be identified and security can be provided against tampering.

Radio frequency identification (RFID) is being used in the IoT since 1999. It is a challenging task to provide security to such a resource-constrained device. In [27] a new secure and lightweight mutual RFID authentication protocol (SecLAP) has been proposed by the authors by removing the vulnerabilities of Lightweight RFID Mutual authentication (LRMI) protocol. The protocol has been proposed for a medical IoT scenario where RFID is used for conveying the patient's information to a cloud. In [28] authors have exploited the vulnerabilities in an existing lightweight authentication protocol for RFID. Authors have presented solutions to improve the security of the existing protocol. In [29] a three way strategy has been used to reduce the footprints of RSA public key cryptography. Machine learning and parallel processing have been used to identify the anomaly and implement RSA on sensors. Another Lightweight Anonymous Authentication Protocol (LAAP) has been proposed in [30] by utilizing one way functions and exclusive-OR operations. LAAP is aimed at edge devices which also have limited resources. It creates small overhead and can be used in both 5G and IoT. In [31] a new set of curves, NUMS, has been used to provide an efficient and fast implementation of ECC. It provides an improved implementation for the asymmetric ciphers. Transport Layer Security protocol (TLS) and IPsec are not suitable for providing security for the constrained devices as the overhead is generated in considerable amount and Internet Key Exchange protocol version 2 (IKEv2) which is used for key establishment is also resource consuming. Shahid Raza and Runar Mar Magnusson have developed TinyIKE [32] which is a lightweight version for the IKEv2. In [33] authors have presented an extensive evaluation of cooperative communication for the physical layer security that can be utilized in resource-constrained devices.

Though, a number of surveys including various LWC algorithms have been reported in the literature. But most of them are focused on single category of cipher. Few of them are presenting all type of LWC ciphers, but an in depth analysis has not been carried out. The work presented in [13, 14, 18, 19, 21] is not the current state-of-art. To the best of our knowledge the work presented in this paper is an extensive overview of all the crucial LWC primitives till 2019. In this paper 21 lightweight block ciphers (LWBC), 19 lightweight stream ciphers (LWSC), 9 lightweight hash functions (LWHF) and 5 variants of elliptic curve cryptography (ECC) has been discussed i.e. in total 54 LWC primitives are compared in their respective classes. Comparative analysis of some new LWC primitives such as QTL, Fruit-v2, Espresso, Lizard, SFN and Neeva have also been included in the present work and the same has not been done in any of the existing literature till date.

Our contribution can be summarized as follows:

- Our work covers a wide range of LWC primitives, symmetric, and asymmetric algorithms along with lightweight hash functions covering current state of the art till date. Some newly proposed ciphers, SFN, ANU, Neeva, Fruit-v2, Espresso, Lizard etc., have also been analyzed in the present work which has not been included in any of the existing surveys.
- A comparative analysis of the ciphers has been carried out in terms of chip area, energy and power, efficiency (hardware and software), throughput, latency and FoM.
- To the best of our knowledge comparative analysis based on the FoM has not been carried out in the available literature. (FoM is an important parameter which helps in making the evaluation of throughput independent of CMOS technology)
- Research gaps are identified are open research issues which are required to be addressed.

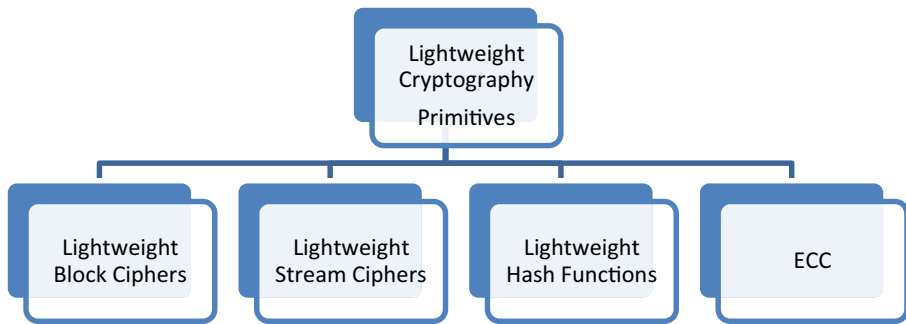


Fig. 2 Lightweight cryptographic primitives for IoT

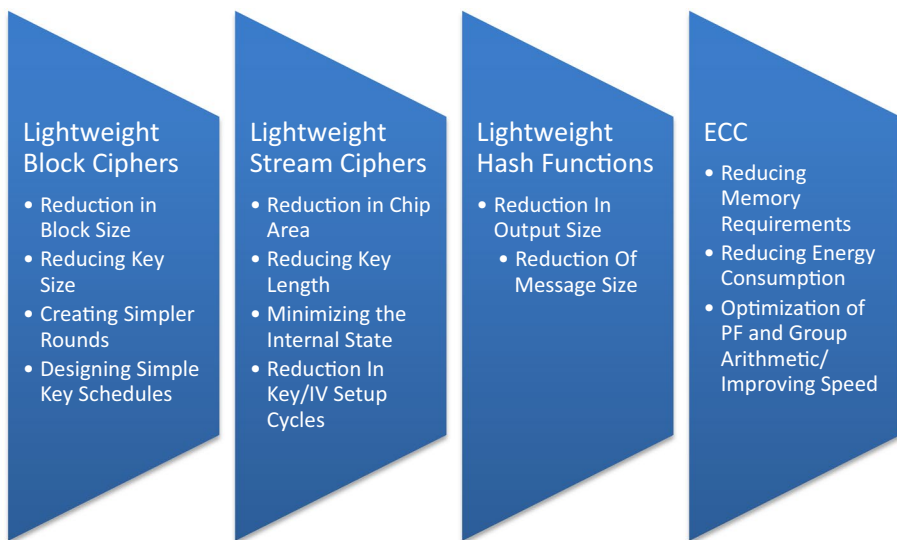


Fig. 3 Research gaps for LWC primitives

3 Classification and Application of Lightweight Cryptographic Primitives

There are mainly four types of lightweight cryptographic primitives that are available for the use. As presented in Fig. 2 the lightweight cryptography primitives can be classified as Lightweight Block Cipher (LWBC), Lightweight Stream Ciphers (LWSC), Lightweight Hash Functions (LWHF) and Elliptic Curve Cryptography (ECC) [13]. The factors on which the lightweight cryptographic primitives can be analyzed are key size, block size, number of rounds, and structures. ECC is another option for lightweight cryptography [34]. Being an asymmetric cipher, it can provide authentication and non-repudiation as well (Fig. 3).

An important question is how to decide whether a given primitive or algorithm is lightweight or not? Device capabilities, hardware implementation, and software implementation of the algorithm are three qualities used to categorize the cryptographic

algorithm. Table 1 shows the three categories of lightweight cryptographic algorithms, low-cost cryptographic algorithm, and ultra-lightweight cryptographic algorithm. The first characteristic is device capability which relates to the resource-constrained nature of the device like 8051 and ATtiny 45. These two devices are more constrained in resources in comparison of devices based on ATmega128 etc. Hardware-based algorithm implementation provides details of the chip area or Gate Equivalence (GE) that is required for the implementation of the algorithm. It also reports the complexity in terms of a number of logic gates. Finally, software implementation helps in classification of cipher based on RAM and Read-Only Memory (ROM) requirements for its implementation [19]. Table 1 also shows the ciphers which are placed in a particular category based upon these three factors. This classification means that one cannot use a LWC primitive requiring higher resources on a device of lesser capacity.

IoT architecture plays an important role in deciding the security requirements for an IoT system or applications [18]. Most of the developers consider two types of architecture for IoT; one is three-layered and another five-layered. Every layer has its security requirements. Security can be implemented at application layer, adaptation layer, transport layer, network layer, and physical layer. The physical layer is also known as perception layer. There is a vast scope for providing security in IoT. Hardware insecurity, lightweight cryptographic algorithms, lightweight trust management system, lightweight secure routing protocols, lightweight antimalware solutions, physical wireless insecurity, Distributed Denial of Service (DDoS) attacks, and privacy protection issues are the areas where one can carry out research work [35].

In the IoT, each layer must have a unique mechanism for security. Table 2 shows different security protocols used at various layers. It also shows the corresponding ciphers that are used in these protocols. Ciphers that have been utilized for the security purpose are SNOW-3G for universal mobile telecommunication system (UMTS), ZUC for UMTS, A5/1 in general packet radio services (GPRS), CLEFIA in ISO/IEC 29192 and KASUMI in global system for mobile communication (GSM), UMTS and GPRS. Constrained Application Protocol (CoAP) is an application layer protocol designed for IoT. It utilizes advanced encryption scheme (AES) [35]. Other protocols that use AES are 802.15.4, Bluetooth Low Energy version 4.2 (BLE 4.2) [36], Wi-Fi protected access version 2 (WPA2) [37], RPL, Internet Protocol version 6 (IPv6), UMTS, 4G, and 6LoWSec etc. From Table 2, it can be observed from this table, AES is the first choice for all the standards. AES has been used for security solutions for all of the layers.

ECC is another primitive that has been used at the physical layer, network layer, and application layer. The protocols which utilize ECC are 6LoWPAN and BLE 4.2. Both AES and digital encryption scheme (DES) could be utilized in IPv6. In IoT, most of the devices at the physical layer, like radio frequency identification (RFID) card or a humidity sensor, have minimal resources in terms of computation, memory, power, and size. On the other hand, traditional cryptographic techniques require higher resources; hence, these techniques are not suited for IoT. As per Padmavathi and Kumari [42], it is not possible to implement the RSA algorithm of 1204 bit in RFID tags. RSA will consume most of the resources of RFID. Hence RFID will not have sufficient resources for other functions. Low-cost implementation of advanced encryption scheme (AES) requires 3600 gate equivalence (GE) as per National Institute of Standards and Technology (NIST). Only PRINT cipher and EPCBC cipher can be used for the RFID having corresponding chip area of 402/726 GE and 1008 GE. Any cipher with more significant gate equivalence will create a problem for the proper functioning of the RFID.

Table 1 Classification of cryptographic algorithms [19]

Category/classification	Device capability	Hardware implementation (chip area or gate equivalence)	Software implementation (ROM/RAM requirements)	Names of ciphers
Ultra lightweight cryptographic algorithms	8051 microcontroller, ATtiny 45	Up to 1000 Gates	4 KB ROM 256 B RAM	QTL, HUMMINGBIRD, Piccolo, Sprout, Fruit-v2, KATAN, KATANTAN,
Low-Cost Cryptographic algorithms	ATmega 128	Up to 2000 gates	4 KB ROM 8 KB RAM	PRESENT, MIBS, SIMON, TWINE, Grain, Grain-128, Midori, Trivium, WG-8, Espresso, Lizard, ECC
Lightweight cryptographic algorithms	Rest above	Up to 3000 gates	32 KB ROM 8 KB RAM	CLEFIA, DEXL, SOSEMANUK, ECC

Table 2 Application of ciphers for security

Cipher (mechanism)	Layer	Protocol	Technology
AES (CCM, CTR,CBC-MAC) [15]	Physical	802.15.4	
AES [35, 36], ECC/ECDH [35]	Physical	BLE 4.2	
AES [35, 37]	Physical (Link layer)	WPA2	802.11
RC4 [35, 37, 44]	Physical (Link layer)	WPA with TKIP	802.11
128-EEA1, 128- EIA1(SNOW-3G)	Physical		UMTS/LTE
128EEA2, 128EIA2(AES)			
128EEA3, 128EIA3 (ZUC) [35]			
KASUMI [35]			GSM, UMTS, GPRS
CLEFIA [19, 43]			ISO/IEC 29192
AES-CCM [41]	Network	RPL	
AES, DES [35]	Network	IPSec, IPv6	
ECC [36, 40]	Network	6LoWPAN	
AES [24]	Adaptation	6LoWPSec	
AES [38, 39], ECC [35]	Application	CoAP	

From the above discussions, it is quite clear that AES and ECC are the primary choice for the security solutions. But researchers need to minimize their implementation costs so that these two can be used in a device as resource deficient as RFID.

4 Lightweight Block Ciphers

Block cipher is a type of symmetric ciphers where a complete block is processed at once. Block ciphers are used for the design of hash functions and message authentication codes (MACs) [44, 45]. Lightweight block ciphers are based on two types of structures: Substitution-permutation network (SPN) and Feistel. Feistel structure uses its round function on only half of the state. It helps in the design of the same circuit for encryption and decryption with minimal overhead. Thus, main advantage of feistel structure is the usage of same program code for the encryption and decryption process. It results in low memory usage. It can be implemented in hardware with low average power. Feistel structure is not suited for the small latency design. SPN will be faster but without a key schedule. Lack of key schedule will make it susceptible to attacks. For the same amount of security margin and same energy expenditure, SPN structure is more suitable as it requires lesser execution round. Under similar conditions, SPN will have lower power expenditure.

Main parameters for the evaluation of a lightweight block cipher are key size, block size, structure type, and the number of rounds. In [46] authors suggest that a lightweight cipher must address the three challenges of minimal silicon area or memory footprint, low power consumption, adequate security level. Cazorla et al. [47] focus on block size, key size, and key scheduling. They insist that to consider an algorithm lightweight it must have a small block size of 32–64 bits as compare to traditional 64 and 128 bits block size. Table 3 presents a comparison of various lightweight block ciphers. It provides details of their structure, key size, block size, rounds needed, and main weaknesses against attacks. Rivest Cipher 4 (RC4) was designed by Ron Rivest [48] in 1987

Table 3 Comparison of lightweight block ciphers

Light weight block cipher	Structure	Rounds	Key size (bits)	Block size (bits)	Weaknesses
RC5 [50], 1994	Feistel	0–255	0–2040 bits	32/64/128	Differential key attacks
TEA [51], 1994	Feistel	64	128		Bad as hash function; related key attacks
XTEA, 1997	Feistel	64	128	64	Related Key rectangle attacks on 36 round
AES [52], 1998	SPN	10, 12, 14	128, 192, 256	128	Bi-clique cryptanalysis
DES [53], 2007	Feistel	16	56	64	–
PRESENT [50], 2007	SPN	31	80/128	64	Side channel attacks, related key attacks on 17 round
CLEFIA [43], 2007	Feistel	2488	128, 192, 256	39/128	Differential fault analysis,
KATAN and KATANAN [55], 2009	NLFSR	254	80	32/48/64	Multidimensional meet in the middle attacks
	NLFSR	254	80	32/48/64	Theoretically broken under the single key setting
MIBS [56], 2009	Feistel with SPN round function	32	64/80	64	Many type of attacks
Humming-Bird [57], 2010	Hybrid	4	256	16	Several attacks
LED [58], 2011	SPN	8 for 64/12 for others	64/80/96/128	64	Bi-clique attacks on reduced rounds, differential fault analysis based on Super-S-box technique
TWINE [59], 2011	GFN Feistel	32	80/128	64	Meet-in-the-middle attacks
KLEIN [60], 2012	SPN	12/16/20	64/80/96	64	Truncated differential attacks
PRINCE [61], 2012	SPN	11	128	64	FX is a questionable choice for new attacks. 12 non-linear layers
ITUBEE [62], 2013	Feistel	–20	80	80	Self-similarity cryptanalysis on 8-round
SIMON and SPECK [63], 2013	Feistel	32–72	64–256	32–128	Attacks on reduced versions and differential fault analysis
	ARX	22–34	64–256	32–128	
RECTANGLE [60], 2014	SPN	25	80/128	64	–
Midori [65], 2015	SPN	16/20	64/128		Many types of attacks
QTL, [66, 67], 2016	Feistel		64/128	64	Susceptible to linear differential attacks
ANU [68], 2016	Feistel	25	80/128	64	–
SNF [69], 2018	Feistel + SPN	32	96	64	–

but it had many vulnerabilities [49]. These vulnerabilities prompted IETF to suggest that cipher should not be used in TLS protocol.

RC5 [50], a fast symmetric block cipher was developed by Rivest in 1994. It offered variable word size with variable key length. It was suitable for hardware and software implementation. Tiny Encryption Algorithm (TEA) [51] was presented in 1994 by Wheeler and Needham, as a lightweight solution for the security. It is a light weight block cipher which is suited for wireless communication. It has a feistel structure. It utilized a 128 bit key for encrypting 64 bits of data. It was quite secure. It used XOR and ADD operations alternatively to provide non-linearity. There were two weaknesses in TEA. First, related-key attacks were possible on TEA, and second key size was 126 not 128 bits. To overcome these two weaknesses of TEA, a new cipher XTEA was purposed in 1996. These two weaknesses were remedied by adjusting the key schedule and slow introduction of key material. In 1998, Joan Daemen and Vincent Rijmen designed the AES [52], a symmetric key cipher. It has 128-bit block that uses SPN structure. It has three key sizes 128 bits, 192 bits, and 256 bits which undergo 10, 12, and 14 rounds respectively. In the list of ciphers other important mentions are Camellia, NOEKEON, and IDEA. These ciphers were an earlier attempt on lightweight block ciphers.

As per [19], the first generation of lightweight cryptography spanned from 2005 to 2012. Some essential efforts during this time are as follows: George Leander et al. [53] presented DESL, a new lightweight version of DES. It is a 64-bit cipher that utilizes 56 bits key and 16 rounds. It is secure against linear and differential cryptanalytic attacks. Bogdanov et al. [54] designed an important block cipher PRESENT to achieve hardware efficiency. Later, this cipher served as the base for many other ciphers. It is a lightweight symmetric cipher with block size of 64-bits and key size of 80 bits. It has very compact area requirements of 1075 GE. Its software implementation was inefficient. While implementing this cipher, one should encrypt a limited amount of data with a given key. C. De Canniere et al. [55] have proposed two ciphers KATAN and KATANANTAN with very compact hardware implementation. In fact, KATANANTAN outperforms KATAN in terms of compactness. MIBS [56] is a 64-bits block cipher that requires 1400 GE on 0.18 μm technology. It is secure against linear and differential cryptanalysis. Daniel Engels et al. [57] have designed Humming Bird in 2010. It was able to achieve larger throughput as compared to PRESENT for size and speed-optimized implementations. It is resistant to linear and differential cryptanalysis. It has a small block size. LED [58] is a block cipher which tried to achieve smallest silicon footprint. Its main design objectives were, compact hardware implementation, ultra-light key schedule, and resistance to related-key attacks and single key attacks. TWINE [59] is a lightweight 64 bits block cipher which utilizes type-2 generalized feistel structure and improved block shuffle. Its most important characteristic is its efficient software implementation on different platforms ranging from 8-bit microcontrollers to high-end CPU. It was proposed by Suzaki et al. in 2011. KLEIN [60] has a compact implementation in hardware. It has good performance even if its software implementation is done on legacy systems. It has a variable key length of 64 bits, 80 bits, and 96 bits. It has a block length of 64 bits and SPN structure. The key schedule is well balanced and secure from related key attacks. Bogroff et al. [61] presented PRINCE in 2012, which provides a new dimension to lightweight cryptography by achieving low latency. It also focuses on hardware implementation. It utilized 128 bits key and was comprised of 64 bits block with 12 rounds. The S-box of this cipher was non-linear i.e. feistel structure. Main advantage of feistel structure is that the same program code can be used for the encryption and decryption process. It also helps in reducing the usage of memory. But the cipher can be susceptible to

related-key attacks if the feistel structure utilizes alternating keys. Some other noteworthy mentions from this generation are Humming-Bird, KASUMI and Piccolo.

The second generation of lightweight cryptography [19] is currently going on. It started in 2013. Karakof et al. [62] have presented a lightweight cipher AKF that is appropriate for RFID tags and WSN etc. It consumes less memory and power. Based on the same lines, they also gave another cipher ITUBEE based on AKF. Ray Beaulieu et al. [63] have shown a simple, lightweight block cipher, SIMON and SPECK in 2015 that performs on heterogeneous platforms with ease due to its inherent simplicity. RECTANGLE [64] was proposed as a block cipher that would be suitable for multiple platforms. Its 80 bit version achieved a chip area of 1600 GE for the hardware implementation while achieving a throughput of 246 kbps at 100 kHz. It utilized a SPN structure.

Midori [65] was presented in 2015. This cipher was optimized for the low energy i.e., energy per bit of encryption. These ciphers consume less than 1.89pJ/bit encryption using STM 90 nm standard library and its version Midori64 and Midori128 outperformed PRINCE and NOEKEON. It is a variant of SPN architecture which utilizes 4 bit S-boxes with 128-bit keys. Midori64 and Midori128 have block size of 64 bits and 128 bits respectively.

QTL, 2016 [66, 67] is an ultra-lightweight block cipher with block size and key size of 64 bits that uses 16 rounds for encryption. It uses a Generalized Feistel Network (GFN) which has the fast diffusion of SP networks. This GFN helps in improving the security of the cipher.

ANU [68], a lightweight block cipher, requires minimal memory size and consumes very less power. It uses 128/80 bit key length. The chip area corresponding to 128 bits key is only 1015 GE. It provides fair security against linear and differential attack, bi-clique cryptanalysis, and zero-correlation attacks.

SFN [69] cipher utilizes the mixed row SPN and Feistel network. The designer has adapted the SP network to work as a feistel network. This modified SPN is used to make similar encryption and decryption while Feistel network is used for key expansion. It uses a key of 96 bits, block of 64 bits, and 32 rounds. Security has been tested against differential and linear attacks, related-key attacks, algebraic attacks, slide attack, meet in the middle attacks, integral attacks, and impossible differential attacks.

5 Lightweight Stream Cipher

Another category of lightweight primitives is stream cipher. In a stream cipher, 'r' bits are encrypted and decrypted at a time. Table 4 gives the comparison table for the lightweight stream ciphers based on key size, chip area, IV, IS, throughput. It also mentions the type of registers in their design along with the CMOS technology used in the implementation. In the first generation of the lightweight cryptography, famous lightweight stream ciphers were A5/1, Rabbit, Grain, and Trivium. A5/1 [70], which was presented in 1987, was used for GSM. Rabbit [71] was presented in 2003. It was one of the earlier efforts on lightweight cryptography. It occupied 3800 GE on 0.18 μm CMOS technology.

The important lightweight stream ciphers are Grain, Mickey and Trivium these are finalists to e-STREAM project phase 3. Grain [72] is a stream cipher which utilizes two registers with a non-linear output function. It can be used for limited memory, gate count, and power consumption. Its key size is 80 bits. Trivium [73] is a synchronous stream cipher that was inspired by block ciphers. It has very efficient hardware implementation.

Grain128-a is a version of Grain [74]. It is a lightweight stream cipher which has a built-in support for authentication. It utilizes the different non-linear functions to provide security against the known attacks. It uses a key of 128 bits. It also has variable tag sizes.

Salsa-20 [75] and Sosemanuk [76] are the finalists of the eSTREAM profile. Salsa-20 was presented by D.J. Berenstein in 2005. It has two key sizes of 128 bits, and 256 bits. But its hardware implementation required a chip area of 12126 GE. The structure of Salsa-20 was Add-Rotate-XOR, which worked on 32 bits word. The initialization vector (IV) and internal state (IS) were 128 bits and 512 bits respectively. Sosemanuk structure was based on linear feedback shift register (LFSR) and FSM. Its key size lies between 128 bits and 256 bits. IV has a value of 64 bits and 128 bits corresponding to the key sizes. Its implementation for 128 bits and 256 bits required 2700 and 4100 GE. Other important ciphers of this generation are MICKEY [77], Chacha [78] Encoro 80 [79], Encoro 128 [80], SNOW-3G [81], A2U2 [82], and Quavium [83]. SNOW-3G was an important cipher that was utilized for UMTS. The internal state of SNOW is 608 bits.

In the second generation of lightweight cryptography, famous stream ciphers are WG-8, Fruit, Plantlet, Espresso, and Lizard. WG-8 [46] is a lightweight stream cipher which utilizes low energy. It belongs to Welch-Gong stream cipher family and is resistant to many frequent attacks against stream ciphers. Sprout cipher [84] was proposed in 2015. Fruit cipher [85] was based on Sprout which was not secure, so designers improved it. Fruit structure is utilized on LFSR and Non-Linear Feedback Shift Register (NLFSR). Its internal state for LFSR was 43 bits while that of NLFSR is 37 bits. Fruit is an ultra-lightweight cipher. Plantlet [86] is an ultra-lightweight cipher occupying only 928 GE. It was presented in 2016. But its security was compromised. It is based on LFSR and NLFSR with a counter register. Espresso [87] is designed for 5G application and is based on Galois NLFSR structure. Finally, the latest one is LIZARD [88] it has low-cost implementation and occupies 1161 GE with 120 bits key-size and 64 bits IV. The structure is NLFSR.

6 Lightweight Hash Functions

Lightweight hash functions are another way to provide security. They create a fixed-length ‘message digest’ from an arbitrary-length message. This ‘message digest’ is used to ensure the integrity of the transmitted data. The footprint of a hash function is determined by the number of state bits, and the size of functional and control logic used in the round function. Some important hash functions are as follows: Jean-Philippe Aumasson et al. [89] in 2010 have presented a new hash family Quark, which is inspired by stream cipher Grain and block cipher KATAN. It utilizes sponge construction and composed of three instances, namely U-Quark, d-Quark, and s-Quark. U-Quark and s-Quark provides 64-bit and 112-bit security. Power consumption for these two is 2.44 μ Watts and 4.35 μ Watts, while chip area requirements are 1379 GE and 2296 GE respectively. Lesamanta-LW [90] was proposed by in 2010. It is a 256-bit hash function.

It performs well on 8-bit CPU as well as high-end server. Kavun and Yalcin [91] presented Keccak in 2010. It has a ‘message digest’ of 160 bits, but chip area for implementation is a bit larger than lightweight standards. It has a maximum chip area of 4763 GE for parallel Keccak implementation and 2079 GE for serial Keccak implementation while the minimum size for parallel implementation is 409 GE and 252 GE for serial Keccak. PHOTON [92] is the most lightweight family of hash functions. Designers have used sponge functions for domain extension algorithms and AES like internal permutation. They have

Table 4 Comparison of lightweight stream ciphers

Light weight stream cipher	Key size	Area (GE)	IV	Type	Internal state (IS)	Throughput (Mbps)	CMOS process
A5/1, 1987 [70]	64	923	22	LFSR		0.050	
Rabbit 2003 [71]	128	3800 4100	64	Chaotic Table + simple arithmetic	513 bit	0.080	180 nm
Grain [72], 2005	80, 128	1294 3239	64, 96	LFSR, NFSR	–	724.6 9876.5	130 nm
Trivium [73], 2005	80	2580 4921	80	3SHR	–	327.9 22299.6	130 nm
Salsa 20/r 2005, [75]	128, 256	12126	128	ARX	512 bit	990	130 nm
Grain128a [74], 2006	128	1857 4617	96	LFSR + NLFSR		925.9 14,479.6	130 nm
Sosenanuk 2008 [76]	128, 256	4100	64, 128	LFSR + FSM	–	800	0.09 μ m/90 nm
MICKEY [77], 2008	80, 128	3188 5039	0–80, 0–128	Galois LFSR + NLFSR	–	454.5	130 nm
CHACHA 2008 [78]	256	750	128	ARX			
Encoro 128 2009 [80]	128,	4100	64,	PRNG		800	90 nm
Encoro80 2008 [79]	80	2700	64				90 nm
SNOW-3G 2010 [81]	128	–	128	LFSR + FSM	608 bit		
A2U2 2011 [82]	56	500 284		LFSR + 2NLFSR		0.050	
Quavium 2012 [83]	80	3496 2372(3-round version)	80	4 Trivium like SHR	288 bit		
WG-8 [46], 2013	80	1786 3942	80	LFSR + WG		500 6710	65 nm
Sprout 2015 [84]		813 839		NLFSR + LFSR + Counter Reg		0.100	180 nm 90 nm
Fruit-v2 2016 [85]	64/80	990	64	LFSR + NLFSR	LFSR43 + NFSR37	0.100	90 nm
Plantlet 2016 [86]	80	928	90	LFSR + NLFSR + Counter			180 nm
Espresso 2017 [87]	128	1500	96	Galois structure NLFSR			90 nm

Table 4 (continued)

Light weight stream cipher	Key size	Area (GE)	IV	Type	Internal state (IS)	Throughput (Mbps)	CMOS process
Lizard 2017 [88]	120	1161	64 3578	NLFSR	121		180 nm

also introduced a new mixing layer building method because of which authors were able to achieve the 1120 GE of area requirement and 64-bit collision-resistant security. It is a family of hash functions, which is defined as PHOTON- $n/s/s'$ where n is the output size, s is input rate and s' is output rate. SPONGENT [93] uses the simplest round function, which helps in achieving the compact design. It has five variants with size 88–256 bits which utilize a chip area of 738–1950 GE. It also provides flexibility in terms of serialization and speed. Table 4 presents crucial lightweight hash functions. GLUON [94] is another lightweight hash that was presented by Berger et al. It is based on filtered Feedback with a Carry Shift Register (FCSR) and utilizes sponge function. It generates a message digest of 160 bits and implementation occupies 2799 GE. GLUON provides 80-bit security. L-Hash [95] can generate a message digest of 80 bits, 96 bits, and 128 bits. It provides 60-bit collision-free security. It uses Feistel-PG structure, which improves diffusion speed. Its implementation requires 817–1028 GE of chip area, which comes from the linear diffusion layer and S-boxes that are hardware friendly.

Hash-One [96] produces a message digest of 160 bits and provides a collision resistance security of minimum 80 bits and pre-image resistance of 160 bits. It uses two NFSR for the sponge state updating. Chip area for implementation is merely 1006 GE.

Neeva [97] is the latest of these lightweight hash functions. It can be used for RFID technology. It is based on sponge structure and provides collision resistance of 112 bits. It is faster than Spongnet-224.

7 ECC for IoT Security

Asymmetric ciphers are also used by security providers. These ciphers provide authentication as well as confidentiality. Larger key size and higher memory consumption are two factors that make asymmetric ciphers a lesser choice. In public-key cryptosystem, RSA and ECC are two primitives that can be used for this purpose. ECC provides the same level of security with smaller key size as compared to RSA. ECC with El-Gamal and Diffie-Hellman key exchange algorithm can be utilized for IoT security. On the basis of execution time, AES [20] is 100–1000 times faster than ECC on 8-bit microcontrollers. Execution time can be improved by reducing the computational complexity of the algorithm. Yet ECC remains the popular choice of the designers to provide security in the IoT scenario. It can also be verified from Table 2.

ECC [34] was introduced by Miller and Neal Koblitz in 1985 independently. These elliptic curves were defined over a finite field. These fields follow specific rules and have a cardinality ' p ' which represents the number of elements in the field. ECC provided better security with smaller key size. A general Koblitz curve can be shown as:

$$E(F_p) : y^2 = x^3 + ax + b \mod p \quad (1)$$

Here $a, b \in F_p$ and.

It means all the elements and operations are performed with the modulo- p and lies in the field.

As the security of RSA algorithm lies in computational hardness of the discrete logarithm problem. In ECC, discrete logarithm problem changes to elliptic curve discrete logarithm problem (ECDLP) and multiplicative groups are replaced by different groups without any compromise in computational hardness. In these groups, multiplication and squaring are replaced with addition and doubling operations which help in fast

execution. Choice of curve models also helps in faster and efficient implementation of scalar multiplications. The specifications and models of such curves have been provided by the NIST and Safecurves website [98]. Safecurves also provides the criteria to select the more secure curves. Montgomery and twisted curves are examples of such curves. Equation (2) gives montgomery curves, while Eq. (3) provides twisted curves:

$$E_{\text{Mon}} : By^2 = x^3 + Ax^2 + x \quad (2)$$

where $A, B \in F_p$ and $\text{Char}(F_p) \neq 2, 3$ and $(A^2 - 4)B \neq 0$

Montgomery curves [99] presented in Eq. (2) were introduced in 1987 by Peter Montgomery. These are a special type of elliptic curves which help in efficient implementation of ‘variable base scalar multiplications’. Twisted curves [100] as mentioned in Eq. (3) were presented by D.J. Bernstein in 2008. These curves are better suited for the ‘faster-fixed-base scalar multiplication’.

$$E_{\text{TW}} : ax^2 + y^2 = 1 + dx^2y^2 \quad (3)$$

where $a, d \in F_p$ and $ad(a - d) \neq 0$.

One important requirement of the secure elliptic curves as per Safecurves [98] is that curve must have a large prime factor of at least 200-bit. P-224 and P-256 are some other curves which have designs elliptic curves using standardized NIST primes. Many variants of ECC are proposed to make it suitable for the resource- constrained devices of WSN and IoT. Some of these attempts are summarised in the Table 5.

Table 5 presents a comparison of different versions of lightweight implementations of ECC. It provides the details of the operations used for the faster execution and the devices used for the implementation. In [105–107] it is shown that the ECC can be utilized to protect data in IoT. In [105] the authors used ECC to provide security in Smart Grid. It provided mutual authentication without revealing the identity of the smart meter with lesser communication costs. Zhe Liu et al. [106] has emphasized that ECC can be utilized as Lightweight Cryptographic primitive. They achieved lesser computation time for 80–128 bit security levels. This implementation was resistant against timing and SPA attacks. Twisted Edward curves performed better in memory and power consumption as compared to standard models of ECC. Tseng et al. [107] have used ECC for the protection of medical data using dynamic elliptic curve cryptosystem (Table 6).

8 Discussion and Future Directions

With the massive data connectivity, IoT has increased the challenges for security. Resource-constrained devices like RFID and sensor motes contradict with the resource requirements of available security solutions. It has also provided impetus to the research in the field of lightweight cryptography. Lightweight cryptography has emerged as a potential solution for the security by providing low cost, low latency implementation that consumes smaller chip area, and lesser memory consumption i.e. RAM or ROM.

Security, chip area, power and energy consumption, FoM, latency are the parameters that help in the evaluation of a cipher. Based on these parameters, different ciphers in their respective category are compared.

8.1 Security

Security provided by a cipher is the primary factor to evaluate a cipher. Security of a cipher can be evaluated on its resistance against various types of attacks. Security is the logarithmic measure of the fastest known computational attacks a cipher. Its unit of measure is 'bit.' Maximum protection provided by a symmetric cipher can be the length of its key. For asymmetric ciphers, it is less than half of its key length. In Fig. 4, the graph compares the security provided by various lightweight hash functions. Spongent delivers the highest security for bit collision that is 128 bits. Lesmanta-LW also provides the 128-bit security. Quark and Neeva both provide 112 bits of security. Security provided by a cipher also gives an idea about the time that would be needed to break a cipher with available resources.

8.2 Chip Area

Second most important parameter of evaluation is the chip area required for the implementation of the cipher. It is related to hardware implementation. Computational complexity and chip area occupied during the hardware implementation is measured in the gate equivalence (GE). It is the ratio of layout area of application measured in μm^2 and the corresponding area of NAND 2 gate. It depends on the CMOS technology. CMOS technology is also essential in the hardware implementation of the ciphers. It affects both gate equivalence and energy consumption. Chip area is a crucial parameter and must have a small value. Figure 5 shows the GE required by different lightweight stream ciphers. It can be seen that A5/1, A2U2, Sprout, Fruit-v2, and Plantlet are the ciphers that lie in the category of ultra-lightweight, i.e. it requires a chip area less than 1000 GE.

While Lizard, espresso, WG-8, Grain and Grain-128 can be categorised as 'low cost' as their chip area lies between 1000 and 2000 GE. Trivium, Sosemanuk, Encoro-80 and Quavium are classified in 'lightweight' as their chip area is less than 3000 GE. Mickey though misses the mark for LWC but can be considered for the difference is marginal.

Figures 6, 7, and 8 provides a comparison between the GE occupied by LWBC and LWHF. Among LWHFs KECCAK, Spongent, L-Hash and Hash-one come under Ultra-lightweight while GLUON is a lightweight cipher and Quark, PHOTON can be put in Low-Cost category. In block ciphers, QTL and LED are two ciphers that come under

Table 5 Comparison of lightweight hash functions

Lightweight hash function	Size (bits)	Gate equivalent (GE)	Security
Quark, 2010 [89]	128–224	1379/2296	64/112 bit security
Lesamanta-LW, 2010 [90]	256	8240	120–128 bit security
KECCAK, 2010 [91]	160	4763–252	
PHOTON, 2011 [92]	80–256	1120	64-bit equivalent security
SPONGENT, 2011 [93]	88–256	738–1950	40–128 bit security
GLUON, 2012 [94]	160	2799	80-bit security
L-Hash, 2013 [95]	96	817/1028	60-bit security
Hash-One, 2016 [96]	160	1006	80 bit security/160 bit preimage resistance/80 bit collision resistance
NEEVA, 2016 [97]	224	–	112-bit security

the ultra-lightweight primitives while all others shown in Fig. 6 are in low-cost category. Detail of the classification of the ciphers is provided in Table 1. Chip area is a measure of the computational complexity of the ciphers. It also affects the power and energy consumption of the cipher and the throughput provided by it.

CMOS technology is also an essential part of the evaluation criteria as it affects the other metrics of performances as well. One cannot compare two ciphers without considering their CMOS technology. As CMOS technology also decides the chip area required in cipher's implementation. Lesser chip area would be occupied if the CMOS technology is 18 nm as compared to 45 nm. It will lead to lesser power consumption. In [19] it has been shown that gate density (kGE/mm²) becomes 800 from 6 when technology used changes from 35 μ m to 0.065 μ m while power (nW/MHz/GE) goes down from 18 to 5.68. It clearly points out that it is unfair to compare two ciphers implemented on two different CMOS technologies. To make this point more clear one can refer to Figs. 6 and 7 in which two different implementations correspond to different chip areas occupied by the LED cipher. Figure 9 shows the details of CMOS technology used for the implementation for the lightweight block ciphers in Fig. 7.

Most of the lightweight block ciphers considered in our survey are implemented on 0.13 μ m. TEA and XTEA which belong to the first generation are implemented on 0.35 μ m. while the later on like TWINE and its variants are implemented on 0.09 μ m.

8.3 Throughput

Throughput gives the number of bits generated in one second at a specific frequency during encryption and decryption procedure of the cipher. Reported throughput is projected at this specified frequency. This frequency is specified either 100 kHz in case of hardware implementation or 4 MHz for software implementation. Throughput can be calculated by the help of Eq. (4) given below:

$$T = \frac{B \times F}{N} \quad (4)$$

where T is the throughput; B is size of data in bits for that is processed in one encryption or decryption or block size and F stands for frequency, while N denotes the number of cycles per block. In this case, frequency and cycles per block vary from one processor to another. Figures 10, 11, and 12 show the throughput provided by the LWSC, LWBC, and LWHF respectively.

Grain-128 provides the highest throughput followed by Sosemanuk and Encoro-80. Other ciphers have comparatively lesser throughput.

Figures 11 and 12 compares the throughput of the different LWB ciphers at 100 kHz frequency. PRINCE and KLEIN provide the highest throughput. DESL comes after them while Rectangle, Piccolo-80 and PRESENT come thereafter.

8.4 Latency

Real time IoT applications like ITS and Unmanned Aerial Vehicles (UAVs) both requires security solutions with minimum time lag. Such applications make 'latency' a critical parameter as both latency and security of information are critical requirements. Latency can be

Table 6 A comparison of various ECC variants

ECC variant	Key operation	Hardware and software used
Nano ECC [34]	Tate pairing and assembler routines	Tiny OS, Labview, Mica2, Tmote Sky
Micro ECC [101]	Multi-precision addition and subtractions, Jacobian projective co-ordinates, 16/32 bit data path, additional memories, Modular multiplication algorithm	Xilinx Virtex-II Pro, Microsemi Smart fusion
Tiny ECC [102]	Projective co-ordinate systems, barret reductions, hybrid multiplication, sliding window method and shamir's trick	MICAz, Telos, Tmote, Imote
WM-ECC [103]	Modified Jacobian co-ordinates, long division, great divide scheme, sliding window method, Shamir's trick	MICAz, Telos/Tmote
MoTE ECC [104]	Optimal prime fields, fixed base comb method, ladder method	MICAz

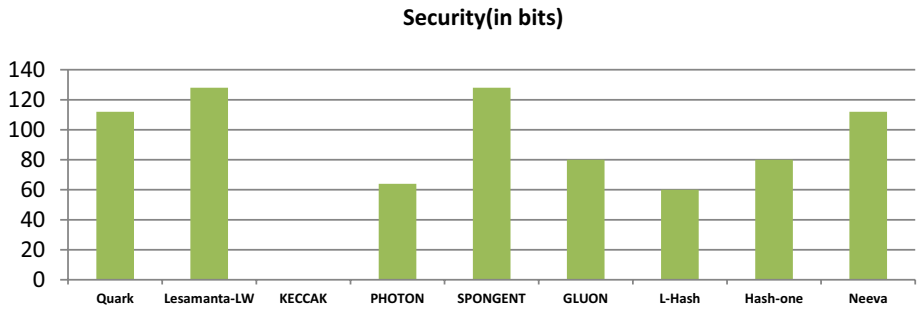


Fig. 4 Equivalent security (in bits) provided by LWHF

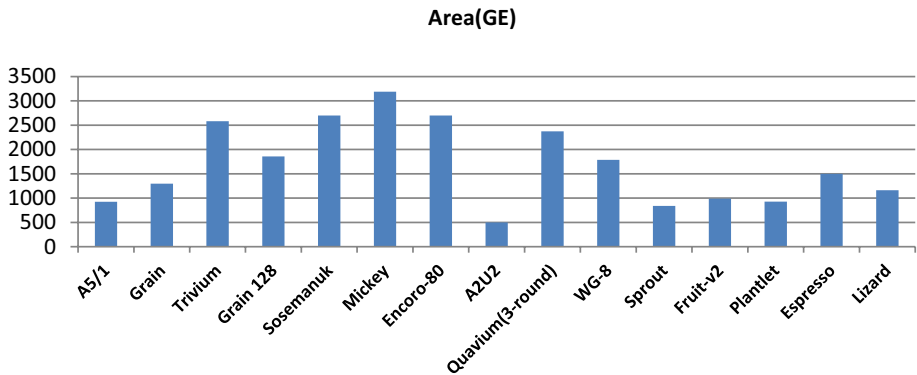


Fig. 5 Gate Equivalence for the lightweight stream cipher as per Table 4

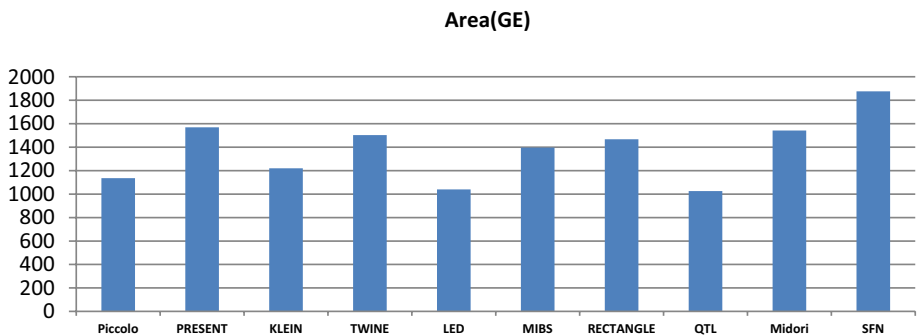


Fig. 6 Gate equivalence for the lightweight block cipher as per Table 3

defined as the time taken for the computation of one block of either plain-text or cipher-text [19, 61, 108]. Here, computation of one block of plain-text stands for decryption and computation of one block of cipher-text stands for the encryption process.

$$L = k \times t_{\text{cycle}} \quad (5)$$

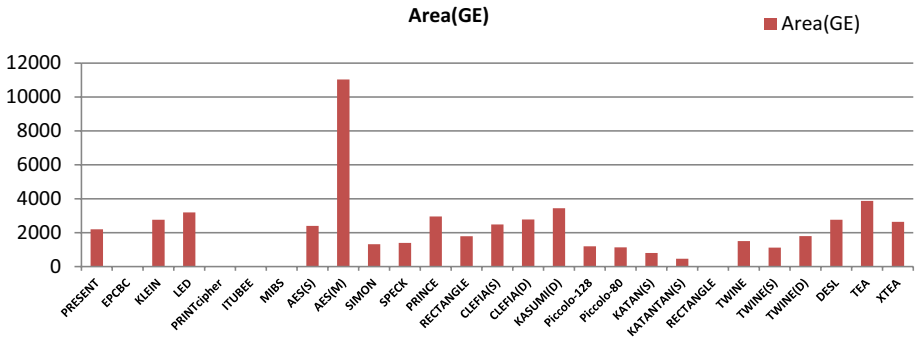


Fig. 7 Gate equivalence for various block ciphers

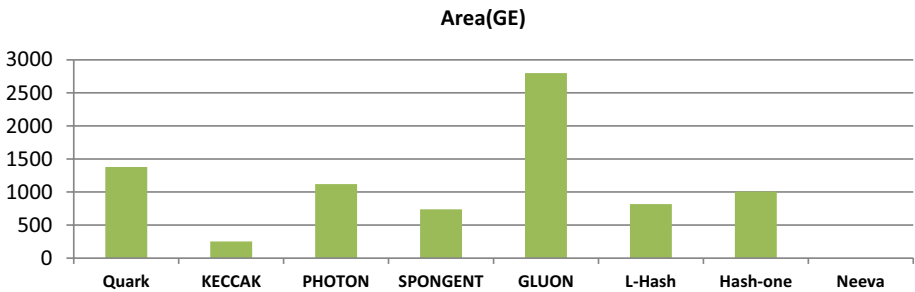


Fig. 8 Gate equivalent for lightweight hash functions

L stands for latency, K is the number of clock cycles required to compute one block of cipher-text, t_{cycle} means the time of one cycle.

Normally block ciphers are preferred in context of latency. Figure 13 shows the latency of different lightweight block ciphers. Prince cipher has been especially designed for the low latency requirements. Similarly, from the Fig. 13, it is clear that RECTANGLE, TWINE, KLEIN, DESL, and PRESENT are some other low latency ciphers.

8.5 Power and Energy

Power and Energy consumption is an important parameter. Power can be estimated based on GE and corresponding CMOS technology [17, 19]. With a change in the CMOS technology, gate density also changes. If CMOS technology changes from μm scale to nano-meter (nm) scale gate density increases. As a result, power expenditure per MHz per GE reduces by a factor between 2 and 3. Energy expenditure per bit can be calculated by the relation presented in Eq. (6) as:

$$E_b = \frac{L \times P}{B} \quad (6)$$

where E_b is energy expenditure per bit, which is measured in micro joule (μJ). L is latency, which is measured in time required for one block encryption or number of clock cycles required for one block of encryption. P is power consumed by hardware/software

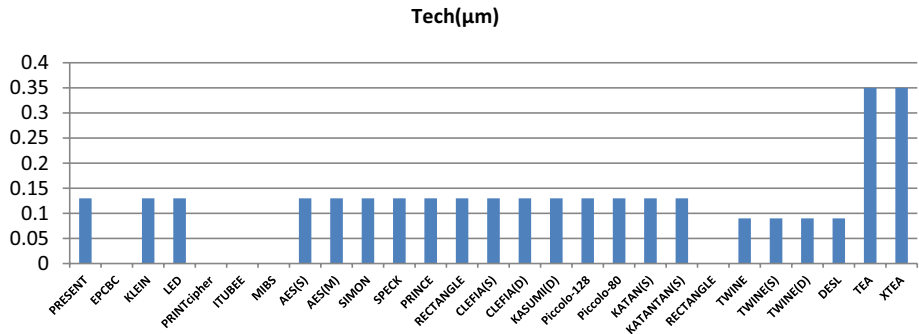


Fig. 9 Tech (μm) of implementation of evaluated ciphers Table 3

implementation that is measured in micro Watt (μW). B stands for the block, as mentioned above.

In Fig. 14, AES (M) consumes maximum power hence it should be least preferred for the resource-constrained devices. TEA and XTEA ciphers are other expensive choices in terms of power consumption. KATAN, KATAN(TAN), Piccolo-80/128, all versions of TWINE cipher, DESL, PRESENT and AES (S) are preferable choices when looked upon the power requirements. Other reasonable choices can be CLEFIA, RECTANGLE, SPECK LED and KLEIN (Figs. 15, 16, and 17).

In context of energy consumption the effectiveness changes a bit due to consideration of block size and latency. PRINCE, RECTANGLE, KLEIN, TWINE, PRESENT Piccolo-80/128, DESL and LED are the ciphers that perform better than other ciphers. AES (S) remains better than AES (M). But the worst performers are TEA and XTEA. AES (S) competes with other ciphers in this category as well though performance is not that better.

8.6 Hardware and Software Efficiency

Hardware and Software Efficiency are another important parameter which can be used to evaluate a cipher. It is the ratio of the performance of cipher with the cost incurred. Performance is measured in 'Throughput' and 'cost' is given by the 'chip area' required for the hardware implementation of the cipher. Hardware efficiency should be as high as

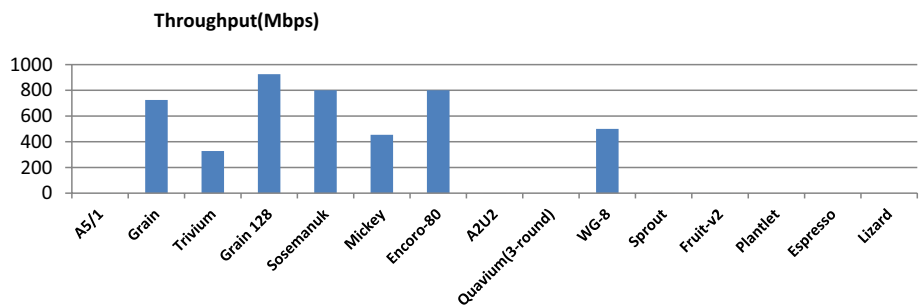


Fig. 10 Throughput for the LWSC

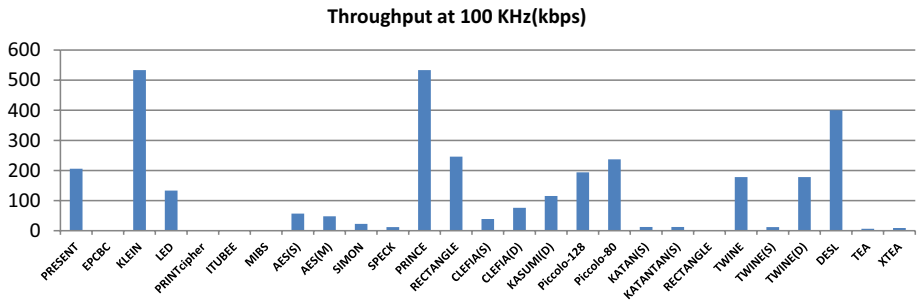


Fig. 11 Throughput of the LWBC

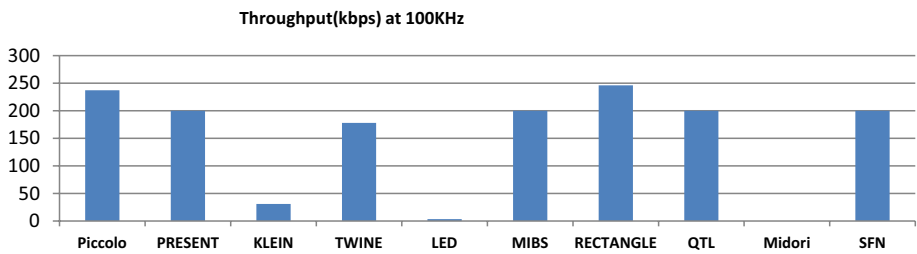


Fig. 12 Throughput for the LWBC

possible. For the hardware implementation, this hardware efficiency [13], [19] is calculated as per Eq. (7) presented below:

$$HE = \frac{T}{C} \quad (7)$$

where HE stands for Hardware Efficiency, T for throughput that is achieved at a particular frequency, mostly 100 kHz, to be measured in kbps here, and C is complexity which is to be measured in the KGE.

Piccolo-80 outperforms every other cipher in this context. It is followed by KLEIN and PRINCE. Piccolo-128, RECTANGLE, TWINE and PRESENT are other important performers under this parameter in descending order of the performance.

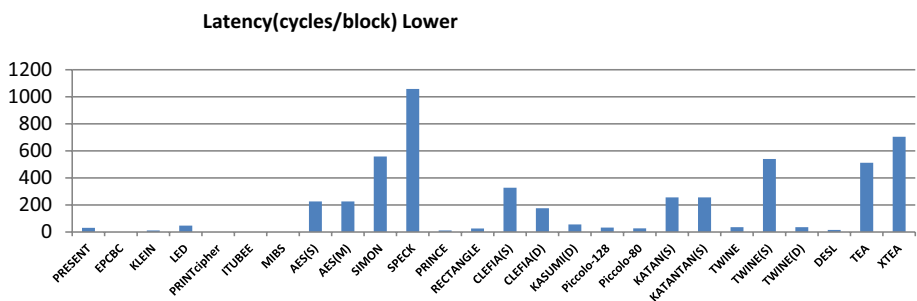


Fig. 13 Latency of different lightweight block ciphers

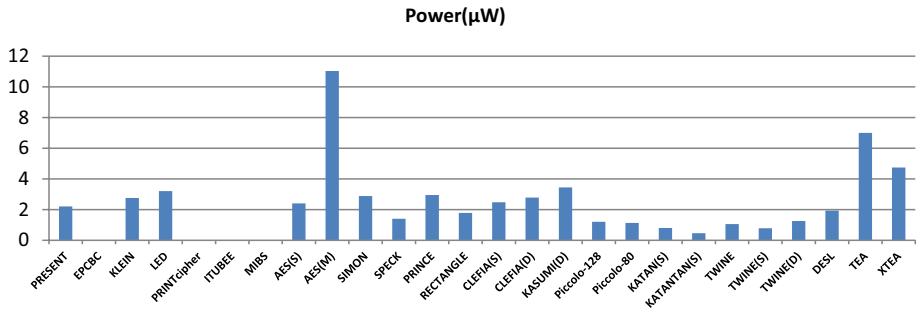


Fig. 14 Power consumption by lightweight block ciphers [19]

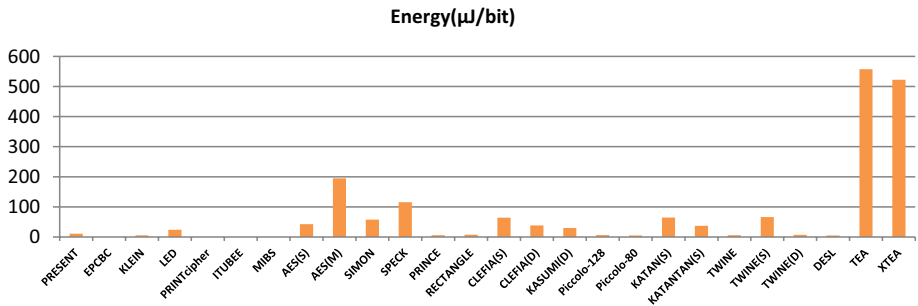


Fig. 15 Energy consumption by lightweight block ciphers [19]

On the other hand, software efficiency [13, 19] can be calculated with the help of Eq. (8) as follows:

$$SE = \frac{T}{CS} \quad (8)$$

where SE is mnemonic for the Software Efficiency, T stands for throughput which is to be measured in kbps achieved at a fixed frequency, mostly 4 MHz. CS stands for Code Size, which is executable code measured in kilobytes (KB) which is an estimate of ROM memory.

8.7 Figure of Merit

Another metric, which was presented by Badel et al. [17], is the figure of merit (FoM). It was presented considering the weakness of the efficiency metric for hardware implementation. Power consumption also depends on the technology chosen and the method of simulation as mentioned in the parameter ‘chip area’. FoM is independent of the process involved and includes the influence of power as well. Using Eq. (9) FoM can be calculated as:

$$FoM = \frac{T}{A^2} \quad (9)$$

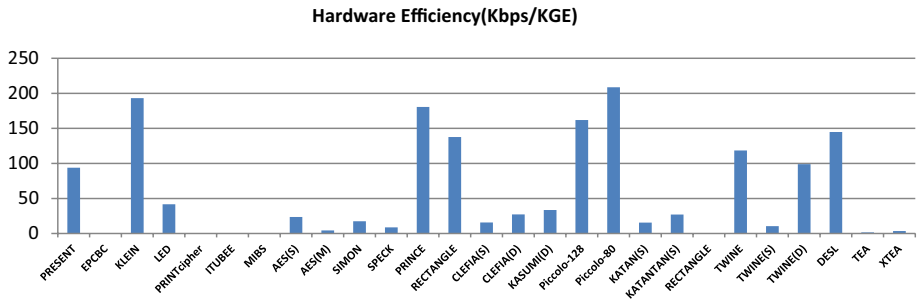


Fig. 16 Hardware efficiency of the LWBC

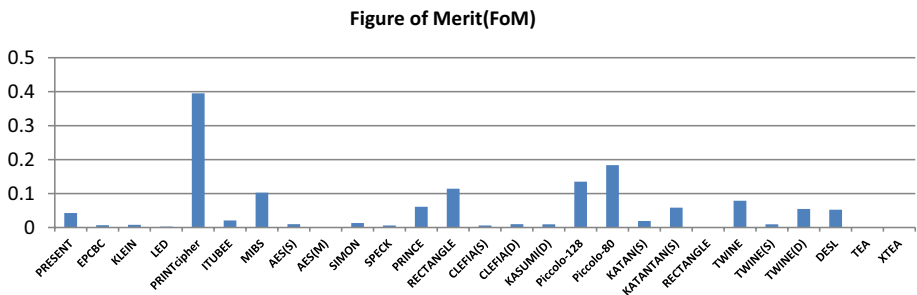


Fig. 17 Figure of merit of LWBC

Here T is throughput, and A is the area of implementation which is measured in GE. Based on these factors, one can evaluate the available ciphers and can decide about the trade-off one has to make while selecting the cipher.

PRINT cipher outperforms all other ciphers on this parameter. Piccolo-80 remains the second best in terms of FoM.

In the current scenario, many lightweight ciphers are available, as mentioned in section II, yet AES is mostly used in different standards for security. A simple calculation reveals that with the current computational capabilities and optimistic assumptions [20], it would take at least 22 million billion years to break a 126 bit AES- key. It means that the level of security provided by the AES is quite good. AES is far ahead of any other competitor. It outperforms many other contenders in security and chip area. While in all other parameters it is notable performer. Its performance in all of the parameters justifies its wide usage. But AES might not prove a good option for an RFID device. AES can consume all available resources of such a resource-constrained device. It clearly points out for the development of lightweight solutions for the IoT. The logarithmic measure of the fastest known computational attack provides the cryptographic strength. Cryptographic strength of most of the symmetric ciphers is equal to their respective key length. For asymmetric ciphers, it falls below half of key length. But asymmetric ciphers can provide confidentiality, authentication, and non-repudiation. ECC has cryptanalytic strength nearly half of its key length, which makes it suitable for security in IoT. ECC can be used for the authentication of devices and access control mechanisms. It can be more useful for IoT where sensors

are deployed in an open environment. Already ECC has been utilized in standards used for the IoT. More than four thousand publications on Science-Direct suggest that researchers have used it widely to device security solutions for constrained devices.

8.8 Research Gaps

Although many new lightweight ciphers have been proposed but there is scope of improvement in terms of latency reduction, security improvement, minimizing internal states, reducing energy and power consumption, overhead reduction, reduction in chip area etc. Ever evolving new attacks also pose a threat to level of security provided by the ciphers. Figure 3 presents the research gaps that can be used to develop new lightweight algorithms. It shows the focus area for each of the primitives separately.

Lightweight block ciphers are facing challenges in:

- Reduction of block size.
- Reducing the key size without compromising the security.
- Creating simpler rounds.
- Designing simple key schedules.

Challenges before Lightweight stream ciphers can be summarized as follows:

- Reduction in chip area.
- Reduction of key length.
- Minimizing the internal state.
- Reduction in number of key/IV setup cycles.

Lightweight Hash functions can work on:

- Reduction in output size and,
- Reduction in message size.

Developers of ECC can focus on the following issues:

- Reducing the memory requirements.
- Reducing the energy consumptions.
- Optimization of Prime Fields (PF) and Group arithmetic.
- Improving the execution speed.

Currently, ECC lags behind AES in term of speed while it does conform to the standards of ultra-lightweight cryptography. It is essential for the ECC designers to reduce the memory requirements, so that it may require lesser RAM and ROM for its operations. If ECC meets these challenges, it can become the first choice of security solutions.

9 Conclusion

Lightweight and low cost cryptographic algorithms are being developed for resource constrained IoT settings. These are evaluated on the basis of chip area occupied in hardware or memory requirements for their software implementation. IoT applications such as ITS and UAV require the information security at very low latency. This need has added a new dimension in the design of lightweight cryptography. Latency and chip area have become two important design parameters in the era of IoT. A cipher can be evaluated on the basis of security, chip area, throughput, latency, hardware and software efficiency, and figure of merit. Prince and Klein ciphers can be used for the low latency requirements. Despite of the continuous development of ciphers, AES remains the preferred choice for provisioning of the security. Based on the extensive evaluation of the ciphers in this work, it is established that AES is the most competitive cipher among block ciphers. In asymmetric cryptography, ECC remains the important option which provides authentication and non-repudiation in addition to the confidentiality. Ever evolving attacks underline the need for development of new lightweight ciphers. In future, researchers can focus on smaller key or block size, simpler rounds and key schedules in the development of LWBC. While in the field of LWSC, minimization of key length, internal state and initialization vector would be prime objectives. For LWHF designer must try to reduce message and output size while providing high bit security. It is important for a LWC to occupy small chip area for its implementation. For asymmetric ciphers, design of optimized prime fields and group arithmetic can help in faster execution and reduced memory requirement.

References

1. Evans, D. (2011). *The Internet of Things: How the next evolution of Internet is changing everything*”, CISCO, San Jose, CA, USA, white paper, 2011. https://www.cisco.com/c/dam/en_us/about/ac79/docs/innov/IoT_IBSG_0411FINAL.pdf.
2. Dhanda, S. S., Singh, B., & Jindal, P. (2019). Wireless technologies in IoT: Research challenges. In K. Ray, S. Sharan, S. Rawat, S. Jain, S. Srivastava, & A. Bandopadhyay (Eds.), *Engineering vibration, communication and information processing*. Lecture Notes in Electrical Engineering, Vol. 478. Springer, Singapore.
3. Internet Security Threat Report. (2019). Vol. 24, Symantec.
4. Ronen, E., & Shamir, A. (2016). Extended functionality attacks on IoT devices: The case of smart lights. In *Proceedings of the 2016 IEEE European symposium on security and privacy (SP'16)*, March 2016 (pp. 3–12).
5. Michele, B., & Karpow, A. (2014). Watch and be watched: Compromising all Smart TV generations. In *Proceedings of the 2014 IEEE 11th Consumer Communications and Networking Conference (CCNC'14)*. IEEE, 2014 (pp. 351–356).
6. Bachy, Y., Basse, F., Nicomette, V., Alata, E., Kaaniche, M., Courrage, J. C., & Lukjanenko, P. (2015). Smart-TV security analysis: practical experiments. In *Proceedings of the 2015 45th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN'15)*. IEEE, 2015 (pp. 497–504).
7. Unuchek, R. (2016). *Obad.a Trojan Now Being Distributed via Mobile Botnets*. Retrieved September 14, 2016, <https://securelist.com/blog/mobile/57453/obad-a-trojan-now-being-distributed-via-mobile-botnets/>.
8. Dhanjani, N. (2017). *Hacking lightbulbs*. Retrieved April 12, 2017, <http://goo.gl/Ry2521>.
9. Notra, S., Siddiqi, M., Gharakheili, H., Sivaraman, V., & Boreli, R. (2014). An experimental study of security and privacy risks with emerging household appliances. In *Proceedings of the 2014 IEEE Conference on Communications and Network Security (CNS'14)*. IEEE, 2014 (pp. 79–84).
10. Ge, M., Hong, J. B., Alzaid, H., & Kim, D. S. (2017). *Security modeling and analysis of cross-protocol IoT devices*. IEEE Trustcom/BigDataSE/ICSS (pp. 1043–1048).
11. http://www.owsap.org/index.php/OWASP_Internet_of_Things_Project.

12. Kamal, R. (2017). *Internet of Things: Architecture and Design Principles*, (p. 403), TMH, India, ISBN-13: 978-93-5260-522-4.
13. Singh, S., Sharma, P. K., Moon, S. Y., & Park, J. H. (2017). Advanced lightweight encryption algorithms for IoT devices: Survey, challenges and solutions. *Journal of Ambient Intelligence & Human Computing*. <https://doi.org/10.1007/s12652-017-0494-4>
14. Biswas, K., Muthukkumarasamy, V., Wu, X. W., & Singh, K. (2016). Performance evaluation of block ciphers for wireless sensor networks. In R. Choudhary, J. Mandal, N. Auluck, & H. Nagarajaram (Eds.), *Advanced Computing and Communication Technologies. Advances in Intelligent Systems and Computing*, Vol. 452. Springer, Singapore.
15. Granjal, J., Monteiro, E., & Silva, J. S. (2015). Security in the integration of low-power wireless sensor networks with the internet: A survey. *Ad Hoc Networks*, 24, 264–287.
16. Zhao, K., & Ge, L. (2013). A survey on the internet of things security. In: *2013 9th International Conference on Computational Intelligence and Security (CIS)*, IEEE (pp. 663–667).
17. Badel, S., Dağtekin, N., Nakahara, J. J., Ouafi, K., Reffé, N., Sepehrdad, P., & Vaudenay, S. (2010). ARMADILLO: A multi-purpose cryptographic primitive dedicated to hardware. In: *Proceeding of International Workshop on Cryptographic Hardware and Embedded Systems* (pp. 398–412). Berlin: Springer.
18. Ammar, M., Russello, G., & Crispo, B. (2018). Internet of Things: A survey on the security of IoT frameworks. *Journal of Information Security and Applications*, 38, 8–27.
19. Hatzivallis, G., Fysarakis, K., Papaefstathiou, I., & Manifavas, C. (2018). A review of lightweight block ciphers. *Journal of Cryptographic Engineering*, 8, 141–184.
20. Schinianakis, D. (2017). Alternative security options in the 5G and IoT Era. *IEEE Circuits and Systems Magazine, Fourth Quarter* (pp. 6–28).
21. Kong, J. H., Ang, L.-M., & Hatzivallis, K. (2015). A comprehensive survey of modern symmetric cryptographic solutions for resource constrained environments. *Journal of Network and Computer Applications*, 49, 15–50.
22. Mohd, B. J., Hayajneh, T., & Vasilakos, A. V. (2015). A survey on lightweight block ciphers for low-resource devices: Comparative study and open issues. *Journal of Network and Computer Applications*, 58, 73–93.
23. Aziz, A., & Singh, K. (2018). Lightweight security scheme for Internet of Things. *Wireless Personal Communication Issue: 104*, 2/2019, Springer online available: 26 Oct 2018. <https://doi.org/10.1007/s11277-018-6035-4>.
24. Meddeb, A., & Glissa, G. (2019). 6LoWPSec: An end-to-end security protocol for 6LoWPAN. *Ad Hoc Networks*, 82, 100–112. <https://doi.org/10.1016/j.adhoc.2018.01.013>.
25. Wu, X.-W., Yang, E.-H., & Wang, J. (2017). Lightweight security protocols for Internet of Things. *IEEE Conference*.
26. Schinianakis, D. (2019) Lightweight security for the Internet of Things: A soft introduction to physical unclonable functions. *IEEE Potentials*, March/April 2019 (pp. 21–28). Doi: <https://doi.org/10.1109/MPOT.2018.2849850>. Date of publication: 6 March 2019.
27. Aghili, S. F., Mala, H., Kaliyar, P., & Conti, M. (2019). SecLAP: Secure and lightweight RFID authentication protocol for Medical IoT. *Future Generation Computer Systems*, 101, 621–634. Doi: <https://doi.org/10.1016/j.future.2019.07.004>.
28. Wang, K.-H., Chen, C.-M., Fang, W., & Tsu-Yang, W. (2018). On the security of a new ultra-lightweight authentication protocol in IoT environment for RFID tags. *Journal of Supercomputing*, 74, 65–70. <https://doi.org/10.1007/s11227-017-2105-8>.
29. Domb, M. (2017). An adaptive lightweight security framework suited for IoT. In J. Sen (Ed.), *Internet of Things: Technology, Applications and Standardization*, IntechOpen. <http://dx.doi.org/10.5772/intechopen.73712>.
30. Gope, P. (2019). LAAP: Lightweight anonymous authentication protocol for D2D-Aided fog computing paradigm. *Computers & Security*, 86, 223–237. <https://doi.org/10.1016/j.cose.2019.06.003>.
31. Liu, Z., & Seo, H. (2019). IoT NUMS: Evaluating NUMS elliptic curve cryptography for IoT platforms. *IEEE Transactions on Information Forensics and Security*, 14, 3.
32. Raza, S., & Magnusson, R. M. (2019). TinyIKE: Lightweight IKEv2 for Internet of Things. *IEEE Internet of Things Journal*, 6(1), 856–866.
33. Pahuja, S., & Jindal, P. (2019). *Cooperative communication in physical layer security: Technologies and challenges, wireless personal communication*. Berlin: Springer Nature.
34. Szczechowiak, P., Oliveira, L. B., Scott, M., Collier, M., & Dahab, R. (2008) NanoECC: Testing the limits of elliptic curve cryptography in sensor networks. In *Wireless Sensor Networks—EWSN 2008*, Vol. 4913 of Lecture Notes in Computer Science. (pp. 305–320). Berlin: Springer Verlag.

35. Frustaci, M., Pace, P., Aloï, G., & Fortino, G. (2018). Evaluating critical security issues of the IoT world: Present and future challenges. *IEEE Internet Of Things Journal*, 5(4), 2483–2495.
36. Chakrabarty, S., & Engels, D. W. (2016). Black networks for Bluetooth low energy. In *Proceedings of the IEEE International Conference Consum. Electron. (ICCE)*, Las Vegas, NV, USA (pp. 11–14).
37. Adnan, A. H., et al. (2015). A comparative study of WLAN security protocols: WPA, WPA2. In *Proceedings of the International Conference on Advances in Electrical Engineering (ICAEE)*, Dhaka, Bangladesh, 2015 (pp. 165–169).
38. Dierks, T., & Rescorla, E. (2004). *The Transport Layer Security (TLS) Protocol Version 1.1*, RFC4346, 2006.
39. McGrew, D., & Bailey, D. (2012) *AES-CCM Cipher Suites for Transport Layer Security (TLS)*, RFC 6655.
40. Blake-Wilson, S., Bolyard, N., Gupta, V., Hawk, V., & Moeller, B. (2006). *Elliptic Curve Cryptography (ECC) Cipher Suites for Transport Layer Security (TLS)*, RFC 4492.
41. Thubert, P., et al. (2012). *RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks*, RFC 6550.
42. Padmavathi, B., & Kumari, S. R. (2013). A survey on performance analysis of DES, AES and RSA algorithm along with LSB substitution. *International Journal of Science and Research*, 2(4), 170–174.
43. Shirai, T., Shibutani, K., Akishita, T., Moriai, S., Iwata, T. (2007). The 128-bit blockcipher CLEFIA (extended abstract). In: *Fast Software Encryption (FSE 2007)*, Springer, LNCS, 4593 (pp. 181–195).
44. Poonam, J., & Brahmjit, S. (2017). Optimization of the security-performance tradeoff in RC4 encryption algorithm. *Wireless Personal Communications*, 92(3), 1221–1250.
45. Poonam, J., & Brahmjit, S. (2017). Security-performance tradeoffs in a class of wireless network scenarios. *Journal of Networks and System Managements*, 25(1), 83–121.
46. Fan, X., Mandal, K., & Gong, G. (2013). Wg-8: A lightweight stream cipher for resource-constrained smart devices. In *International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness* (pp. 617–632). Berlin, Heidelberg: Springer.
47. Cazorla, M., Marquet, K., & Minier, M. (2013). Survey and benchmark of lightweight block ciphers for wireless sensor networks. In: *Proceedings of the SECRIPT*. <http://eprint.iacr.org/2013/295>.
48. Poonam, J., & Brahmjit, S. (2015). Quantitative analysis of the security performance in wireless LANs. *Journal of King Saud University-Computer and Information Sciences*, 29(3), 246–268.
49. Poonam, J., & Brahmjit, S. (2015). Experimental study to analyze the security performance in wireless LANs. *Wireless Personal Communications*, 83(3), 2085–2131.
50. Rivest, R. L. (1994). The RC5 encryption algorithm. *Proceeding of international workshop on fast software encryption* (pp. 86–96). Berlin: Springer.
51. Wheeler, D. J., & Needham, R. M. (1994). TEA, a tiny encryption algorithm. *Proceeding of international workshop on fast software encryption* (pp. 363–366). Berlin: Springer.
52. National Institute of Standards and Technology (NIST). (2001). Advanced Encryption Standard (AES). Federal information processing standards publication 197, November 26. <http://csrc.nist.gov/publications/fips/fips197/fips-197.pdf>.
53. Leander, G., Paar, C., Poschmann, A., & Schramm, K. (2007). New lightweight DES variants. In A. Biryukov (Ed.) *The 14th Annual Fast Software Encryption Workshop—FSE 2007*, LNCS 4593 (pp. 196–210). Berlin: Springer-Verlag.
54. Bogdanov, A., Knudsen, L. R., Leander, G., Paar, C., Poschmann, A., Robshaw, M. J. B., Seurin, Y., Vikkelsøe, C.: PRESENT: An ultra-lightweight block cipher. In *Proceeding of Cryptographic Hardware and Embedded Systems—CHES 2007* (pp. 450–466). Springer.
55. De Canniere, C., Dunkelman, O., & Knežević, M. (2009). KATAN and KTANTAN—a family of small and efficient hardware-oriented block ciphers. In *International Workshop on Cryptographic Hardware and Embedded Systems* (pp. 272–288). Springer.
56. Izadi, M., Sadeghiyan, B., Sadeghian, S. S., & Khanooki, H. A. (2009). MIBS: A new lightweight block cipher. In *Proceeding of Cryptography and Network Security-CANS 2009* (pp. 334–348). Springer.
57. Engels, D., Fan, X., Gong, G., Hu, H., Smith, E. M. (2010). Hummingbird: ultra-lightweight cryptography for resource-constrained devices. In *Financial Cryptography and Data Security—FC 2010*, LNCS, 6054 (pp. 3–18). Springer.
58. Guo, J., Peyrin, T., Poschmann, A., & Robshaw, M. (2011). The LED block cipher. In: *Proceeding of Cryptographic Hardware and Embedded Systems-CHES 2011* (pp. 326–341). Springer.
59. Suzaki, T., Minematsu, K., Morioka, S., & Kobayashi, E. (2011) TWINE: A lightweight, versatile block cipher. In *Proceeding of ECRYPT Workshop on Lightweight Cryptography 2011* (pp. 146–169).

60. Gong, Z., Nikova, S., & Law, Y. W. (2012). KLEIN: A new family of lightweight block ciphers. In *Proceeding of RFIDSec 2011*, (pp. 1–18). Springer.
61. Borghoff, J., Canteaut, A., Güneysu, T., Kavun, E. B., Knezevic, M., Knudsen, L. R., Leander, G., Nikov, V., Paar, C., Rechberger, C., Rombouts, P., Thomsen, S. S., Yalçın, T. (2012). PRINCE—A low-latency block cipher for pervasive computing applications. In *Proceeding of ASIACRYPT 2012* (pp. 208–225). Springer.
62. Karakoç, F., Demirci, H., & Harmanç, A. E. (2013). ITUbee: A software oriented lightweight block cipher. In *Proceeding of Lightweight Cryptography for Security and Privacy—LightSec 2013* (pp. 16–27). Springer.
63. Beaulieu, R., Treatman-Clark, S., Shors, D., Weeks, B., Smith, J., & Wingers, L. (2013). The SIMON and SPECK lightweight block ciphers. In *Proceeding of 52nd ACM/EDAC/IEEE, Design Automation Conference (DAC)* (pp. 1–6) IEEE.
64. Zhang, W., Bao, Z., Lin, D., Rijmen, V., Yang, B., & Verbaauwhede, I. (2014). RECTANGLE: A bit-slice ultra-lightweight block cipher suitable for multiple platform. *Science China Information Sciences*, 58(12), 1–15.
65. Banik, S., Bogdanov, A., Isobe, T., Shibutani, K., Hiwatari, H., Akishita, T., et al. (2015). *Midori: A block cipher for low energy* (pp. 411–436). Berlin: Springer.
66. Li, L., Liu, B., & Wang, H. (2016). QTL: A new ultra-lightweight block cipher. *Microprocessors and Microsystems*, 45, 45–55.
67. Sadeghi, S., Bagheri, N., & Abdelraheem, M. A. (2017). Cryptanalysis of QTL Cipher. *Microprocessors and Microsystems*, 52, 34–48.
68. Bansod, G., Patil, A., Sutar, S., & Pisharoty, N. (2016). ANU: An ultra lightweight cipher design for security in IoT. *Security and Communication Networks*, 9, 5238–5251.
69. Li, L., Liu, B., Zhou, Y., & Zou, Y. (2018). SFN: A new lightweight block cipher. *Microprocessors and Microsystems*, 60, 138–150.
70. Biryukov, A., Shamir, A., & Wagner, D. (2001). *Real time cryptanalysis of A5, 1 on a PC, Fast Software Encryption (FSE), LNCS* (Vol. 1978, pp. 1–18). New York: Springer.
71. Boesgaard, M., Vesterager, M., Pedersen, T., Christiansen, J., & Scavenius, O. (2003). *Rabbit: A new high-performance stream cipher, FSE, LNCS* (Vol. 2887, pp. 307–329). Lund: Springer.
72. Hell, M., Johansson, T., & Meier, W. (2005). Grain—A stream cipher for constrained environments. In *Workshop on RFID and Light-Weight Crypto: Workshop Record*, Graz, Austria.
73. De Cannière, C., & Preneel, B. (2005). Trivium—A stream cipher construction inspired by block cipher design principles. *ECRYPT Stream Cipher*. <http://www.ecrypt.eu.org/stream/paper/sdir/2006/021.pdf>.
74. Hell, M., Johansson, T., & Maximov, A. (2006). A stream cipher proposal, Grain-128. In *IEEE International Symposium on Information Theory*, Seattle, WA (pp. 1614–1618).
75. Bernstein, D. J. (2005). The Salsa20 stream cipher, slides of talk. In: *ECRYPT STVL Workshop on Symmetric Key Encryption*. <http://cr.yp.to/talks.html#2005.05.26>.
76. Berbain, C. et al. (2008) Sosemanuk, a fast software-oriented stream cipher. In: M. Robshaw & O. Billet (Eds.), *New Stream Cipher Designs. Lecture Notes in Computer Science*, Vol. 4986. Springer, Berlin.
77. Babbage, S., & Dodd, M. (2008). The MICKEY stream ciphers. In *Proceeding of New Stream Cipher Designs* (pp. 191–209). Berlin: Springer.
78. Bernstein, D. J. (2008). *ChaCha, a variant of Salsa20*. <http://cr.yp.to/papers.html#chacha>. Supersedes: (PDF)2008.01.20.
79. Watanabe, D., Ideguchi, K., Kitahara, J., Muto, K., & Furuichi, H. (2008). Enocoro-80: A hardware oriented stream cipher. In *Third International Conference on Availability, Reliability and Security (ARES 08) 2008*; 1294 (1300): 4–7.49. Systems Development Laboratory, Hitachi.
80. Enocoro-128v2. (2009). *A Hardware Oriented Stream Cipher, Hitachi Ltd*. http://www.hitachi.com/rd/yr/crypto/enocoro/enocoro_spec_20100222.zip.
81. Orhanou, G., Hajji, S. E., & Bentalab, Y. (2010). SNOW 3G stream Cipher operation and complexity study. *Contemporary Engineering Sciences*, 3(3), 97–111.
82. David, M., Ranasinghe, D. C., & Larsen, T. (2011). In *IEEE International Conference on RFID A2U2: A stream cipher for printed electronics RFID tags* (pp. 176–183).
83. Tian, Y., Chen, G., & Li, J. (2012). Quavium—A new stream Cipher Inspired by Trivium. *Journal of Computers*, 7(5), 1278–1284.
84. Armknecht, F., & Mikhalev, V. (2015) On lightweight stream ciphers with shorter internal states. In G. Leander (Ed.) *Fast Software Encryption: 22nd International Workshop, FSE 2015, Istanbul, Turkey, March 8–11, 2015, Revised Selected Papers* (pp. 451–470). Berlin: Springer.<https://doi.org/10.1007/978-3-662-48116-522>.

85. Ghafari, V. A., Hu, H., Xie, C. (2016). *Fruit V2: Ultra-lightweight Stream Cipher with Shorter Internal State*. Cryptology ePrint Archive Report 2016/355. <http://eprint.iacr.org/2016/355>.
86. Mikhalev, V., Armknecht, F., & Muller, C. (2017). On ciphers that continuously access the non-volatile key. *IACR Transmission Symmetric Cryptology*, 2, 52–79. <https://doi.org/10.13154/tosc.v2016.i2.52-79>.
87. Dubrova, E., & Hell, M. (2017). Espresso: A stream cipher for 5G wireless communication systems. *Journal of Cryptography and Communication*, 9(2), 273–289.
88. Hamann, M., Krause, M., & Meier, W. (2017). LIZARD—A lightweight stream cipher for power-constrained devices. *IACR Transmission Symmetric Cryptology*, 1, 45–79. <https://doi.org/10.13154/tosc.v2017.i1.45-79>.
89. Aumasson, J.-P., Henzen, L., Meier, W., & Naya-Plasencia, M. (2010). Quark: A lightweight hash. In *International Workshop on Cryptographic Hardware and Embedded Systems* (pp. 1–15). Springer.
90. Hirose, S., Ideguchi, K., Kuwakado, H., Owada, T., Preneel, B., & Yoshida, H. (2010). A lightweight 256-bit hash function for hardware and low-end devices: Lesamnta-LW. In *Proceeding of International Conference on Information Security and Cryptology* (pp. 151–168). Berlin: Springer.
91. Kavun, E. B., & Yalcin, T. (2010). A lightweight implementation of keccak hash function for radio-frequency identification applications. In *International Workshop on Radio Frequency Identification: Security and Privacy Issues* (pp. 258–269). Springer.
92. Guo, J., Peyrin, T., & Poschmann, A. (2011). *The PHOTON family of lightweight hash functions*, CRYPTO 2011, LNCS 6841, International Association for Cryptologic Research (pp. 222–239).
93. Bogdanov, A., Knežević, M., Leander, G., Toz, D., Varici, K., & Verbauwhede, I. (2011). *SPONGENT: A lightweight hash function*, CHES 2011, LNCS 6917, International Association for Cryptologic Research (pp. 312–325).
94. Berger, T. P., D'Hayer, J., Marquet, K., Minier, M., & Thomas, G. (2012). The GLUON family: A lightweight hash function family based on FCSRs. In A. Mitroksotsa & S. Vaudenay (Eds.) *Progress in Cryptology—AFRICACRYPT 2012*. Lecture Notes in Computer Science, Vol. 7374. Springer, Berlin.
95. Wenling, W., Shuang, W., Zhang, L., Zou, J., & Dong, L. (2013). *LHash: A lightweight hash function (full version)*. <https://eprint.iacr.org/2013/867>.
96. Mukundan, P. M., Manayankath, S., Srinivasan, C., & Sethumadhavan, M. (2016). Hash-One: A lightweight cryptographic hash function. *IET Information Security*, 10(5), 225–231.
97. Bussi, K., Dey, D., Kumar, M., & Dass, B. K. (2016). *Neeva: A Lightweight Hash Function*, IACR Cryptology ePrint Archive, (042). <https://eprint.iacr.org/2016/042>.
98. Bernstein, D. J., & Lange, T. (2014) *SafeCurves: Choosing safe curves for elliptic-curve cryptography*. Retrieved December 1, 2014, from <https://safecurves.cr.yo.to>.
99. Montgomery, P. L. (1987). Speeding the Pollard and elliptic curve methods of factorization. *Mathematics of Computation*, 48(177), 243–264.
100. Bernstein, D. J., Birkner, P., Joye, M., Lange, T., & Peters, C. (2008). Twisted Edwards curves. In *Progress in Cryptology* (pp. 389–405). Berlin: Springer Verlag.
101. Varchola, M., Guneyasu, T., & Mischke, O. (2011). MicroECC: A lightweight reconfigurable elliptic curve crypto-processor. In *The Proceedings of International Conference on Reconfigurable Computing and FPGAs*, 30 November–2 December 2011, Cancun, Mexico. <https://doi.org/10.1109/reconfig.2011.61>.
102. Liu, A., & Ning, P. (2008). TinyECC: A configurable library for elliptic curve cryptography in wireless sensor networks. In *Proceedings of the 7th International Conference on Information Processing in Sensor Networks (IPSN 2008)* (pp. 245–256). IEEE Computer Society Press.
103. Wang, H., & Li, Q. (2008). Efficient implementation of public key cryptosystems on mote sensors. In *Information and Communications Security—ICICS 2006*, Vol. 4307 of Lecture Notes in Computer Science. (pp. 519–528). Berlin: Springer Verlag.
104. Liu, Z., Wenger, E., & Großschädl, J. (2014). MoTE-ECC: Energy-scalable elliptic curve cryptography for wireless sensor networks. In: I. Boureanu, P. Owesarski, S. Vaudenay (Eds.) *Applied cryptography and network security*. ACNS 2014. Lecture Notes in Computer Science, Vol. 8479. Springer, Cham
105. He, D., Wang, H., Khan, M. K., & Wang, L. (2016). Lightweight anonymous key distribution scheme for smart grid using elliptic curve cryptography. *IET Communications*, 14, 1795–1802.
106. Liu, Z., Huang, X., Hu, Z., Khan, M. K., Seo, H., & Zhou, L. (2017). On emerging family of elliptic curves to secure Internet of Things: ECC comes of age. *IEEE Transactions on Dependable and Secure Computing*, 14(3), 237–248.

107. Tseng, C. H., Wang, S.-H., & Tsaur, W.-J. (2015). Hierarchical and dynamic elliptic curve cryptosystem based self-certified public key scheme for medical data protection. *IEEE Transactions on Reliability*, 64(3), 1078–1085.
108. Knezevic, M., Nikov, V., & Rombouts, P. (2012). Low-latency encryption is “Lightweight=Light+Wait”? In E. Prouff & P. Schaumont (Eds.) *CHES 2012, LNCS 7428*, (pp. 426–446).

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